Md Shohanoor Rahman

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Research Interests

- **Focus:** trustworthy and explainable AI, especially testing whether explanations of image classifiers are correct.
- **Methods:** generative models (diffusion, GANs), concept-based explanations, counterfactual evaluation.
- **Areas:** computer vision, medical image analysis (dermoscopy), health data.

Technical Summary

- Research experience in testing concept explanations (TCAV, attribution) against known-target controls.
- Research experience in diffusion counterfactuals: a concept-conditioned DDPM with seven concept knobs.
- Research experience in explainable ML for clinical data with SHAP and LIME (two first-author IEEE papers).
- Research experience in GAN-based synthetic tabular data and its evaluation in eight dimensions.
- Hands-on experience in PyTorch training of CNNs, ViTs and diffusion models on multi-GPU and TPU.
- Hands-on experience in rigorous evaluation: positive and null controls, rules fixed in code before each run.
- Hands-on experience in production software engineering (3.5+ years): Spring Boot, Kafka, Kubernetes, Azure.

Education

Rajshahi University of Engineering & Technology (RUET) — Rajshahi, Bangladesh
B.Sc. in Computer Science & Engineering — *Jan 2017 – Aug 2023*

- **Thesis:** *Generative Adversarial Network based Synthetic Data Generation System* (GitHub).
- **Relevant coursework**

Artificial Intelligence	Applied Statistics and Queuing Theory	Operating Systems
Neural Networks and Fuzzy Systems	Discrete Mathematics	Parallel and Distributed Processing
Data Mining	Numerical Methods	Computer Architecture and Design
Digital Image Processing	Data Structure	Software Engineering
Digital Signal Processing	Computer Algorithms	Information System Analysis and Design
Vector Analysis and Linear Algebra	Database Systems	Technical Writing and Presentation

Publications

Journal manuscript under review

[J1] **Md. Shohanoor Rahman**, Md. Hasibul Islam, Md Kishor Morol, Moinul Hossain, Dip Nandi, Liew Tze Hui. "Testing concept-level audits of melanoma classifiers with diffusion counterfactuals and matched positive controls." Under review at *Scientific Reports*, 2026. Code and result tables: github.com/srahman7900/melanoma-concept-audit.

Peer-reviewed conference papers

[C1] **Md. Shohanoor Rahman**, Abid Hasan Mahmud, S. M. Mehedy Hasan. "Interpretable Multi-Class Obesity Classification Using Bayesian Optimization and SHAP Insights." In *Proc. 28th International Conference on Computer and Information Technology (ICCIT)*, IEEE, Cox's Bazar, Bangladesh, Dec 2025. DOI: 10.1109/ICCIT68739.2025.11491728. *Presented by the author.*

[C2] **Md. Shohanoor Rahman**, Abid Hasan Mahmud, S. M. Mehedy Hasan. "Explainable Machine Learning for Chronic Kidney Disease Prediction with SHAP and LIME." In *Proc. 28th International Conference on Computer and Information Technology (ICCIT)*, IEEE, Cox's Bazar, Bangladesh, Dec 2025. DOI: 10.1109/ICCIT68739.2025.11490146.

Research Experience

Testing concept-level audits of melanoma classifiers [J1] — Remote
First author. Supervisors: Md Kishor Morol, Liew Tze Hui and Dip Nandi — *Dec 2025 – Present*

- Conceived the study, designed the method, wrote all code, ran all experiments on Kaggle (2× T4 GPUs, TPU v5e-8), did the analysis and figures, and wrote the first draft.
- Built the instrument: a concept-conditioned DDPM on ISIC 2020 (128 px) in which each 7-point checklist concept is a knob from 0 to 1. Removed a signal-leak bias in the start noise (cross-validated $R^2 = 0.90$) with colour-calibrated start noise, cutting colour outliers from 78.5–84.8% to 6.6–13.3%.
- Tested the audit, TCAV and observational attribution against classifiers with known targets: 16 matched controls, 12 controls labelled by a second evaluator and 18 permuted-label nulls. Fixed each decision rule in code before its run (SHA-256 fingerprints for the later rules) and reported the first rule, which failed because its quartile controls were confounded.

- The audit found the target in 16 of 16 matched controls and in 12 of 12 second-evaluator controls (three concepts); TCAV found 11 of 16 (own implementation) and 0 of 3 (Captum 0.9.0). The pigmentation sweep raised the melanoma logit in 15 of 15 ISIC 2020 (+3.12) and 6 of 6 HAM10000 classifiers (+4.01), about four times the mean null response.

Explainable machine learning for clinical data [C1, C2] Bangladesh
Group lead. Advisor: S. M. Mahedy Hasan, Assistant Professor, CSE, RUET *Jan 2025 – Dec 2025*

- Led the group while working full time: literature review, all code and both manuscripts. Built SHAP and LIME pipelines for chronic kidney disease prediction (4,921 patient records) and benchmarked eleven classifiers for six-level obesity classification on a new dataset of 650 people collected in Bangladesh.

Undergraduate thesis: GAN-based synthetic tabular data RUET, Rajshahi
With Md. Aukerul Moin Shuvo. Supervisor: Barshon Sen, Assistant Professor, CSE, RUET *Nov 2022*

- Trained CTGAN on a skewed, imbalanced financial dataset (6,819 firms, 95 ratios, 3.2% bankrupt) and evaluated the synthetic data in eight dimensions, from distributions and PCA to detection tests and train-on-synthetic, test-on-real. My part: literature review, exploratory analysis and results analysis.

Professional Experience

Brain Station 23 PLC Dhaka, Bangladesh
Software Engineer *Mar 2023 – Present*

- **IRIS platform, Tennant Company (USA):** Azure telemetry for autonomous cleaning machines; worked on the Overview dashboard, Tech Hub (semantic search, Azure AI Search) and customer onboarding.
- **AB Direct (AB Bank) and NCC Always (NCC Bank):** features for two live internet banking platforms (transfers, bill payment, deposits, service requests).
- **Pocket (ABG Technologies, Bashundhara Group):** admin, customer and merchant portals of a mobile wallet licensed by Bangladesh Bank.

Technical Skills

- **Machine learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, CatBoost; diffusion models (DDPM, DDIM, classifier-free guidance), GANs, concept bottleneck models, CNNs and ViTs.
- **Explainability and statistics:** TCAV (Captum), SHAP, LIME; positive and null controls; Wilcoxon, Benjamini-Hochberg, bootstrap.
- **Programming and tools:** Python, Java, TypeScript, C/C++, SQL; Git, LaTeX, Kaggle (multi-GPU, TPU).
- **Software engineering:** Spring Boot microservices, Angular, Kafka, PostgreSQL, Redis, Kubernetes, Azure.
- **Languages:** Bengali (native), English (professional working proficiency).

Training and Activities

- **DeepLearning.AI (Coursera):** GANs Specialization; four Deep Learning Specialization courses.
- **Projects:** pix2pix (aerial image to map), CycleGAN (image translation), U-Net (biomedical segmentation).
- **RUET Analytical Programming Lab:** two years; programming contests; helped organise a contest.

Test Scores

- **GRE General:** 312 (Quantitative 165, Verbal 147, Analytical Writing 3.0) Jul 2025

References

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